

QUANTUM COMPUTING · BEGINNER SERIES

Check Your Understanding

Test what you have learned about qubits, gates, amplitudes, probabilities, measurement, and interference.

1 What is a qubit?

Which statement best describes a qubit?

- A. A classical bit that can only store 0
- B. A quantum system whose state can be described using amplitudes for $|0\rangle$ and $|1\rangle$
- C. A bit that is always both 0 and 1 when measured
- D. A computer memory cell containing two classical bits

Answer: B

A qubit is a quantum system described by a state such as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

The amplitudes α and β determine the probabilities of obtaining 0 or 1 when the qubit is measured.

What does the X gate do to a computational-basis state?

- A. It changes $|0\rangle$ into $|1\rangle$ and $|1\rangle$ into $|0\rangle$
- B. It always creates a 50–50 superposition
- C. It measures the qubit
- D. It removes the qubit

Answer: A

The X gate behaves like a NOT operation:

$$X|0\rangle = |1\rangle \quad X|1\rangle = |0\rangle$$

So X flips the computational-basis state.

What happens when H is applied to $|0\rangle$?

- A. The state becomes $|1\rangle$
- B. The state remains $|0\rangle$
- C. The state becomes an equal superposition of $|0\rangle$ and $|1\rangle$
- D. The qubit is immediately measured

Answer: C

The Hadamard gate creates an equal superposition:

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

When measured in the computational basis, this gives 0 and 1 with equal probability.

If the state is $|\psi\rangle = \sqrt{3}/2 |0\rangle + 1/2 |1\rangle$, what are the measurement probabilities?

A. $P(0) = 3/4$ and $P(1) = 1/4$

B. $P(0) = \sqrt{3}/2$ and $P(1) = 1/2$

C. $P(0) = 1/2$ and $P(1) = 1/2$

D. $P(0) = 3/2$ and $P(1) = 1/2$

Answer: A

Probabilities are obtained by taking the squared magnitude of each amplitude.

$$P(0) = |\sqrt{3}/2|^2 = 3/4$$

$$P(1) = |1/2|^2 = 1/4$$

Notice that the probabilities add to 1.

A qubit is prepared in $|+\rangle$ and measured in the computational basis. What can the measurement return?

A. Only 0

B. Only 1

C. Either 0 or 1

D. Both 0 and 1 at the same time as the measurement result

Answer: C

The state

$$|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

gives two possible measurement outcomes: **0 with 50% probability** and **1 with 50% probability**. A single measurement produces one definite result.

What is the difference between an amplitude and a probability?

- A. They are exactly the same thing
- B. Amplitudes describe the quantum state, while squared magnitudes of amplitudes give measurement probabilities
- C. Probabilities can be negative
- D. Amplitudes must always be between 0 and 1

Answer: B

Amplitudes are the numbers appearing in the quantum state. Probabilities are calculated from their squared magnitudes:

$$P = |\text{amplitude}|^2$$

Amplitudes may be negative or complex, but probabilities are always real and non-negative.

What does measurement do to a qubit?

- A. It always leaves the qubit in exactly the same superposition
- B. It produces a classical outcome such as 0 or 1
- C. It creates a new qubit
- D. It changes 0 into 1 automatically

Answer: B

When measured in the computational basis, a single qubit produces either 0 or 1. Measurement also projects the quantum state onto the corresponding basis state.

Suppose we run the circuit

$$|0\rangle \rightarrow H \rightarrow M$$

What results should we expect after many runs?

A. Approximately 100% zeros

B. Approximately 100% ones

C. Approximately 50% zeros and 50% ones

D. Exactly 50 zeros and 50 ones regardless of the number of runs

Answer: C

H changes $|0\rangle$ into $|+\rangle$:

$$|0\rangle \rightarrow H \rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$$

Therefore each measurement has a 50% theoretical probability of producing either result. With many shots, the observed counts should become close to 50–50, but they do not have to be exactly equal.

What happens when H is applied twice to $|0\rangle$?

A. The state becomes $|1\rangle$

B. The state becomes $|+\rangle$

C. The state returns to $|0\rangle$

D. The qubit disappears

Answer: C

The first H creates a superposition:

$$|0\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$$

The second H causes the amplitudes to interfere and returns the state to $|0\rangle$.

$$H^2 = I$$

Therefore:

$$|0\rangle \rightarrow H \rightarrow H \rightarrow |0\rangle$$

Why can two H gates undo each other?

- A. Because H is a measurement gate
- B. Because the second H causes interference between amplitudes
- C. Because all quantum gates automatically cancel
- D. Because measurement happens automatically after every gate

Answer: B

The first H creates two amplitudes. The second H transforms those amplitudes again. The positive and negative contributions can interfere constructively or destructively. For $|0\rangle$, the result is:

$$H(H|0\rangle) = |0\rangle$$

This is an important example of **quantum interference**.

★ Reasoning Challenge

Consider the circuit:

$$|0\rangle \rightarrow H \rightarrow H \rightarrow M$$

Before looking at the answer, try to predict the measurement result.

Prediction: The result should always be 0 in the ideal case.

First:

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

Then apply H again:

$$H [(|0\rangle + |1\rangle)/\sqrt{2}] = |0\rangle$$

Therefore the final state is $|0\rangle$.

Key idea:

A quantum circuit is not just about the final probabilities. The amplitudes and their relative signs determine how later gates interfere with one another.

✓ Final Checklist

- ✓ I know that a qubit can be described as $\alpha|0\rangle + \beta|1\rangle$.
- ✓ I know that measurement produces a definite result.
- ✓ I can calculate probabilities from amplitudes.
- ✓ I know what X does to $|0\rangle$ and $|1\rangle$.
- ✓ I know that H creates superposition.
- ✓ I understand that amplitudes can have signs.
- ✓ I understand that repeated measurements reveal probability distributions.
- ✓ I know that $H^2 = I$.
- ✓ I have seen how interference can change the final result.

You now have the basic toolkit for reading simple quantum circuits.

Qubit → State → Gate → Amplitude → Probability → Measurement → Interference